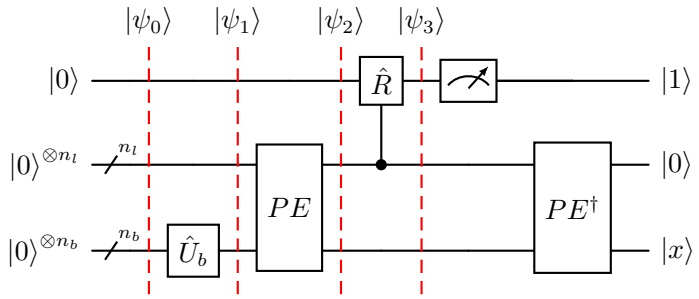

图 6: Circuit of HHL

5.6 Sample-based Quantum Diagonalization of a Chemistry Hamiltonian

Sample-based quantum diagonalization (SQD) algorithm [9, 10] is a method for finding eigenvalues and eigenvectors of quantum opertors, using quantum and distributed classical computing together. Classical distributed computing is used to process samples obtained from a quantum processor, and to project and diagonalize a target Hamiltonian in a subspace spanned by them. SQD is known to work well when the target eigenstate is sparse.

In 2.3, we have learned in quantum chemistry the formalism of second quantization can be used to describe the electronic structure of a molecule. Upon the definition of annihilation operator and creation operator, we also defined their anti-commutation relations in Eqn. 14. However, in this construction, the spin, which is an important feature of electron, was not taken into account. To compensate that, we introduce lables $\hat{\alpha}_{p\sigma}$, where σ is used to include either spin-up state α or spin-down state β. Accordingly, Eqn 2 is updated into

$$\hat{H} = \sum_{pr\sigma} h_{pr} \hat{a}^\dagger_{p\sigma} \hat{a}_{r\sigma} + \frac{1}{2} \sum_{prqs\sigma\tau} h_{prqs} \hat{a}^\dagger_{p\sigma} \hat{a}^\dagger_{q\tau} \hat{a}_{s\tau} \hat{a}_{r\sigma}, \tag{70}$$

where again h_{pr} and h_{prqs} are complex numbers from molecular integrals.

5.6.1 Unitary Cluster Jastrow (UCJ) ansatz

SQD requires a quantum circuit ansatz to draw samples from. Local UCJ (LUCJ) [11], due to its combination of physics and hardware friendliness, is adopted here. This ansatz is based on UCJ [12] which was first proposed by Robert Jastrow to solve many-body problem with strong forces. To address the short range strong force problem, he introduced a product of exponentialized wavefunctions containing a parameter that can be used in variational computation to reach the minimum energy of a quantum system. This wavefunction, which is later called unitary cluster Jastrow (UCJ), can fit into the current quantum computing to

solve quantum chemistry problem. It looks like

$$|\psi\rangle = \prod_{\mu=1}^{L} e^{\hat{K}_\mu} e^{i\hat{J}_\mu} e^{-\hat{K}_\mu} |\Phi_0\rangle, \tag{71}$$

where $|\Phi_0\rangle$ is a reference state, often taken to be the Hartree-Fock state, and the $\hat{K}_\mu$ and $\hat{J}_\mu$ are defined as [10, 11]

$$\hat{K}_\mu = \sum_{pq,\sigma} K^\mu_{pq} \hat{a}^\dagger_{p\sigma} \hat{a}_{q\sigma},\ \hat{J}_\mu = \sum_{pq,\sigma\tau} J^\mu_{pq,\sigma\tau} \hat{n}_{p\sigma} \hat{n}_{q\tau}, \tag{72}$$

where $\hat{n}_{p\sigma} = \hat{a}^\dagger_{p\sigma} \hat{a}_{p\sigma}$ is known as the number operator and K^μ_{pq} is the complex matrix elements forms an anti-Hermitian matrix, whereas $J^\mu_{pq,\sigma\tau}$ is real and it forms a symmetric matrix. To maintain the spin symmetry, $\hat{J}$ should commute with the z-component of the total spin operator $S_z = \frac{1}{2}\sum_p (\hat{n}_{p\alpha} - \hat{n}_{p\beta})$, which is in essence the weighted summation of particle number. Therefore, the conditions like

$$J^\mu_{pq,\alpha\alpha} = J^\mu_{pq,\beta\beta},\ J^\mu_{pq,\alpha\beta} = J^\mu_{pq,\beta\alpha} \tag{73}$$

should be guaranteed. Meanwhile, note that commutation with S^2 operator requires $J^\mu_{pq,\alpha\alpha} = J^\mu_{pq,\alpha\beta}$, eqn (72) can be reorganized and written as

$$\begin{aligned}\hat{J}_\mu = 2\sum_p J^\mu_{pp,\alpha\beta} \hat{n}_{p\alpha}\hat{n}_{p\beta} + 4\sum_{p<q} J^\mu_{pq,\alpha\beta} \hat{n}_{p\alpha}\hat{n}_{q\beta} + 2\sum_p J^\mu_{pp,\alpha\alpha}[\hat{n}_{p\alpha} + \hat{n}_{p\beta}] \\ + 2\sum_{p<q} J^\mu_{pq,\alpha\alpha}[\hat{n}_{p\alpha}\hat{n}_{q\alpha} + \hat{n}_{p\beta}\hat{n}_{q\beta}]\end{aligned} \tag{74}$$

On the right of the eqn (74), the first term is on-site opposite-spin term; the second one is the non-diagonal opposite-spin term; the third term is the on-site same-spin term, and the last one is the non-diagonal same-spin term.

In eqn (71), the operator $e^{\hat{K}_\mu}$ is also regarded as the rotational operator over the orbitals whose function is to rotate the given basis to new basis. By doing so, the redistribution of the electrons within the orbitals is realized. In this process, the total number of electrons and their spin state are conserved. From the point of view of group theory, the unitary rotational operator $e^{\hat{K}_\mu}$ is a Lie group element while the operator $\hat{K}_\mu$ is its Lie algebra generator. In practice, such a rotation is realized on the quantum computing circuit via a serise of Givens rotation which will be disscussed in details later. In contrast, the number-number interaction operator $\hat{J}_\mu$ realizes the modulation of electron correlation. Therefore, unlike $\hat{K}_\mu$, it is a two-body operator. The sanwitch shape structure $e^{\hat{K}_\mu} e^{i\hat{J}_\mu} e^{-\hat{K}_\mu}$ composed of those two types of operators is a similarity transformation. When it is applied onto a reference state like we saw

in eqn (71), it implies three steps. First, it rotates reference state $|\psi_0\rangle$ into new orbital basis via $e^{-\hat{K}_\mu}$, then $e^{i\hat{J}_\mu}$ will exert electron correlation modulation on the new basis, i.e. the Jastrow factor. After that, it will change the basis back via $e^{\hat{K}_\mu}$.

Now, suppose we are going to use the widely used JW mapping scheme (see section 2.3), N spatial orbitals should be mapped onto a system of $2N$ qubits,

$$\hat{a}^\dagger_{p\alpha} \mapsto (S_+)_p Z_{(p-1)} ... Z_0,\ \hat{a}^\dagger_{p\beta} \mapsto (S_+)_{N+p} Z_{N+p-1} ... Z_0, \tag{75}$$

where $(S_+)_p = (X_p + iY_p)/2$ is the raising operator for qubit p. From here we can see, to achieve linear connectivity, we need $\mathcal{O}(N^2)$ Givens rotations acting on adjacent qubits. Considering eqn (74), the operator $e^{i\hat{J}_\mu}$ is mapped onto a quantum circuit of the form

$$\begin{aligned} e^{i\hat{J}_\mu} \mapsto & \prod_p v_{p,p+N}(2J^\mu_{pp,\alpha\beta}) \prod_{p<q} v_{p,q+N}(4J^\mu_{pq,\alpha\beta}) \prod_p w_p(2J^\mu_{pp,\alpha\alpha}) w_{p+N}(2J^\mu_{pp,\alpha\alpha}) \\ & \times \prod_{p<q} (2J^\mu_{pq,\alpha\beta}) v_{p+N,q+N}(2J^\mu_{pq,\alpha\beta}), \end{aligned} \tag{76}$$

where $w_p(\phi) = e^{-i\frac{\phi}{4}Z_p}$ is a single-qubit gate and

$$v_{p,q}(\phi) = e^{-i\frac{\phi}{4}(Z_p + Z_q - Z_p Z_q)} \tag{77}$$

is a two-qubit gate. Eqn (76) indicates that we need $\mathcal{O}(N^2)$ two-qubit gates with all-to-all connectivity. In summary, we can say the UCJ is not applicabale to current hardware.

5.6.2 Local Unitary Cluster Jastrow (LUCJ)

To overcome the problem of UCJ addressed in the previous subsection, we introduce "local" approximation of the UCJ ansatz, which needs the following modifications for opposite-spin and same-spin number-number terms:

$$\sum_{pq} J_{p\alpha,q\beta} \hat{n}_{p\alpha} \hat{n}_{p\beta} \to \sum_p J_{p\alpha,p\beta} \hat{n}_{p\alpha} \hat{n}_{p\beta} \tag{78}$$

$$\sum_{pq} J_{p\sigma,q\sigma} \hat{n}_{p\sigma} \hat{n}_{q\sigma} \to \sum_p J_{p\sigma,(p+1)\sigma} \hat{n}_{p\sigma} \hat{n}_{(p+1)\sigma},\ \text{where}\, \sigma = \alpha, \beta \tag{79}$$

From the two equations above, we can see the summation over p and q is simplified into summation of p only. By doing so, we neglect all the terms with $p \neq q$ so that only diagonal double occupation penalty terms are kept for opposite-spin terms. As for the same-spin term, the summation is performed over neigboring terms. Obviously, this approximation only takes into account the strongest electronic repulsion. In other words, two sparse matrices with the elements of $J^\mu_{p\alpha,q\alpha}$ and $J^\mu_{p\alpha,q\beta}$ respectively are formed. This allows us to implement the circuit

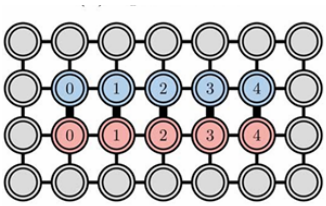

图 7: Square Connectivity

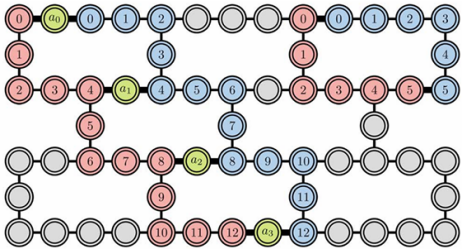

图 8: Heavy-hex lattice

in a constant depth with limited connectivity. For example, if the connectivity is a square lattice show in the figure 7. In vector form, the summation corresponding to eqns (78, 79)

$$\begin{aligned} \boldsymbol{J}^{\alpha\alpha} &: \{(p, p+1), p = 0, ..., N-2\} \\ \boldsymbol{J}^{\alpha\beta} &: \{(p, p), p = 0, ..., N-1\} \end{aligned} \tag{80}$$

Similarly, in the case of the heavy-hex lattice (see fig 8), the sparse matrices are

$$\begin{aligned} \boldsymbol{J}^{\alpha\alpha} &: \{(p, p+1), p = 0, ..., N-2\} \\ \boldsymbol{J}^{\alpha\beta} &: \{(p, p), p = 0, 4, 8, ...(p \leq N-1)\} \end{aligned} \tag{81}$$